

Drones

Definition:

A drone is an unmanned aerial vehicle (UAV), which means it is an aircraft without a pilot. It is controlled by a person or is programmed to fly automatically using programs, sensors and GPS.

Main Parts of a Drone:

- Frame
- Motors
- Propellers
- Electronic Speed Controllers (ESCs)
- Flight Controller
- Sensors
- GPS Module
- Battery
- Remote Controller

- **Frame:**

1. The frame holds all components together.
2. It is the main skeleton of a drone.
3. It supports all the drone's components such as the motors, propellers, battery etc.
4. It provides strength, stability and protection.

● **Motors:**

1. Motors are the muscles of a drone.
2. They convert the Electrical Energy into Mechanical Energy.
3. When the motors spin the propeller rotates and the lift is generated.
4. There are two types of Motors:
 - **Brushed Motors** - They are simple, inexpensive less efficient and has shorter lifespan.
 - **Brushless Motors** - They are more powerful and efficient, have longer lifespan and used for professional purposes.
5. The Brushless Motors are preferred more because of its high speed and greater reliability.

● **Propellers:**

1. Propellers are the rotating blades attached to the motors.
2. They push the air downwards to generate **Lift** allowing the drone to fly.
3. The four propellers balance the force and creates stability.
4. In the drone two propellers go in the clockwise direction and the other two propellers go in the anti clockwise direction.
5. It is designed this way so that it cancels the rotational torque.

- **ESC (Electronic Speed Controller):**

1. ESC is an electronic device that controls the speed of a drone's motor based on the Flight Controller.
2. It controls the motor speed accurately at the required speed.
3. Each motor has its own ESC.
4. The ESC helps to stabilize the drone.

- **Flight Controller:**

1. The Flight Controller is the brain of the drone.
2. It receives the data from sensors and the commands from the pilot then controls the motors through ESC's.
3. It controls the ESC's and the motors.
4. Connects all the major components like sensors, ESC's, GPS and the receiver.

- **Sensors:**

1. Sensors are devices that collect information about the drone and its surroundings and send that information to the flight controller.
2. It measures the speed, height and the direction of the drone.
3. It also detects the drone's position.
4. The common sensors used in drones are
 - Gyroscope - detects rotation
 - Accelerometer - measures acceleration and tilt

- GPS - determines the drone's location
 - Magnetometer - determines the direction
 - Barometer - measures the altitude
5. Sensors are important because they improve navigation accuracy and prevent crashes.

● **GPS Module:**

1. GPS - Global Positioning System
2. GPS is a satellite-based navigation system that helps a drone determine its exact location, speed, and altitude.
3. It helps the drone to follow a planned route.
4. It helps the drone hold its position accurately.

● **Battery:**

1. The Battery is the power source of the drone.
2. It supplies electrical energy to all the drone's components.
3. It powers the motors and propellers and supplies energy to the flight controller and sensors.
4. Most drones use Lithium Polymer (Li-Po) batteries because they are lightweight and provide high current.

● **Remote Controller:**

1. A remote controller is a handheld device used by the pilot to control the drone wirelessly.
2. It sends commands to the drone and controls movement and direction.
3. The Main Controls are:
 - Throttle - Controls altitude
 - Pitch - Moves the drone forward and backwards.
 - Roll - Moves the drone left and right.
 - Yaw - Rotates the drone left or right.
4. It enables safe operation of the drone.

● **Conclusion:**

Drones use components such as motors, propellers, sensors, GPS, and flight controllers to achieve stable flight and perform various tasks. Understanding these components is essential for working with drone technology and related applications.